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A AUROC

A.1 AUROC Algorithm

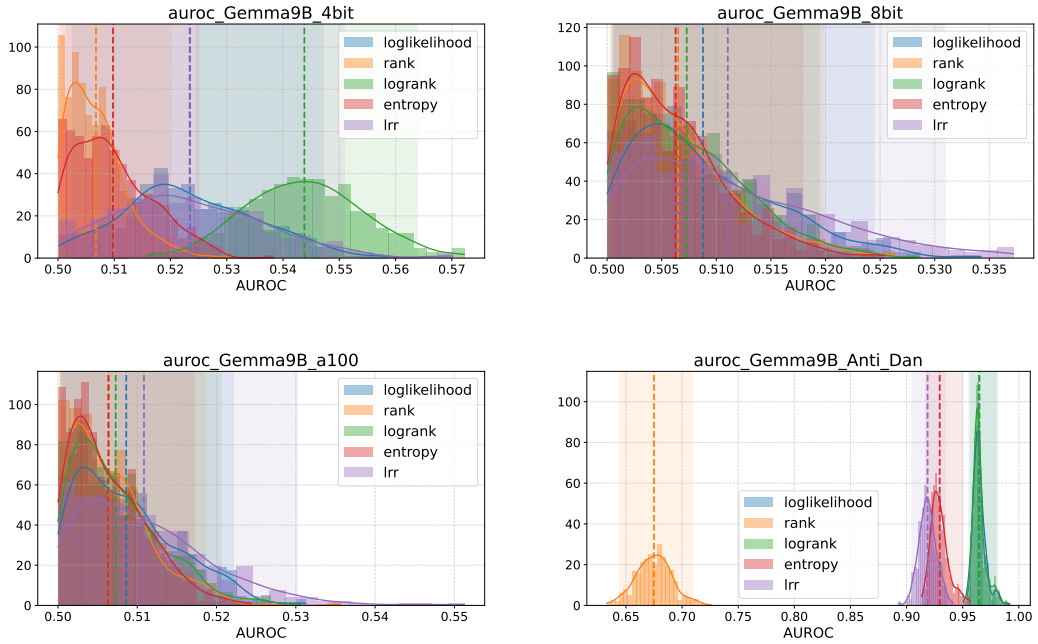
Algorithm 1: Average AUROC for score function evaluation

Input: Prompt set $\mathcal{D}$; models π_{ref}, π_{tgt}; decoding parameters $\varphi = (\tau, L)$, where τ is temperature and L is the maximum generation length; number of prompts n; completions per prompt per model m; score functions $\{\delta_1, \ldots, \delta_K\}$.

Output: Mean AUROC per score function, denoted $\mu_{\text{AUROC}}(\delta)$.

Draw $\{x_1, \ldots, x_n\} \sim \text{Uniform}(\mathcal{D})$;
for $i \in \{1, \ldots, n\}$ **do**
  $\{y_{\text{ref}}^{(j)}\}_{j=1}^{m} \sim \pi_{\text{ref}}(\cdot \mid x_i; \varphi)$;
  $\{y_{\text{tgt}}^{(j)}\}_{j=1}^{m} \sim \pi_{\text{tgt}}(\cdot \mid x_i; \varphi)$;
  $\mathcal{Y}_i \leftarrow \{y_{\text{ref}}^{(j)}\} \cup \{y_{\text{tgt}}^{(j)}\}$;
  $L_i \leftarrow \{0\}^m \cup \{1\}^m$;
  for $\delta \in \{\delta_1, \ldots, \delta_K\}$ **do**
    $S_i \leftarrow \{\delta(y) \mid y \in \mathcal{Y}_i\}$;
    Store $A_i^{\delta} \leftarrow \text{AUROC}(S_i, L_i)$;
for $\delta \in \{\delta_1, \ldots, \delta_K\}$ **do**
  $\mu_{\text{AUROC}}(\delta) \leftarrow \frac{1}{n} \sum_{i=1}^{n} A_i^{\delta}$;

Note. $\text{AUROC}(S, L)$ denotes the standard binary AUROC (Bradley, 1997).

A.2 AUROC Score Distributions

We present the AUROC score distributions from the score function selection experiment described in Section 4.2. Specifically, we evaluated Gemma-2-9B-it, LLaMA-3.2-3B-Instruct, and Mistral-7B-Instruct, and visualized the distributions when distinguishing the original model outputs from three types of variants: (1) quantized versions, (2) models subjected to jailbreaking prompts, and (3) models served by A100 or external API providers.